

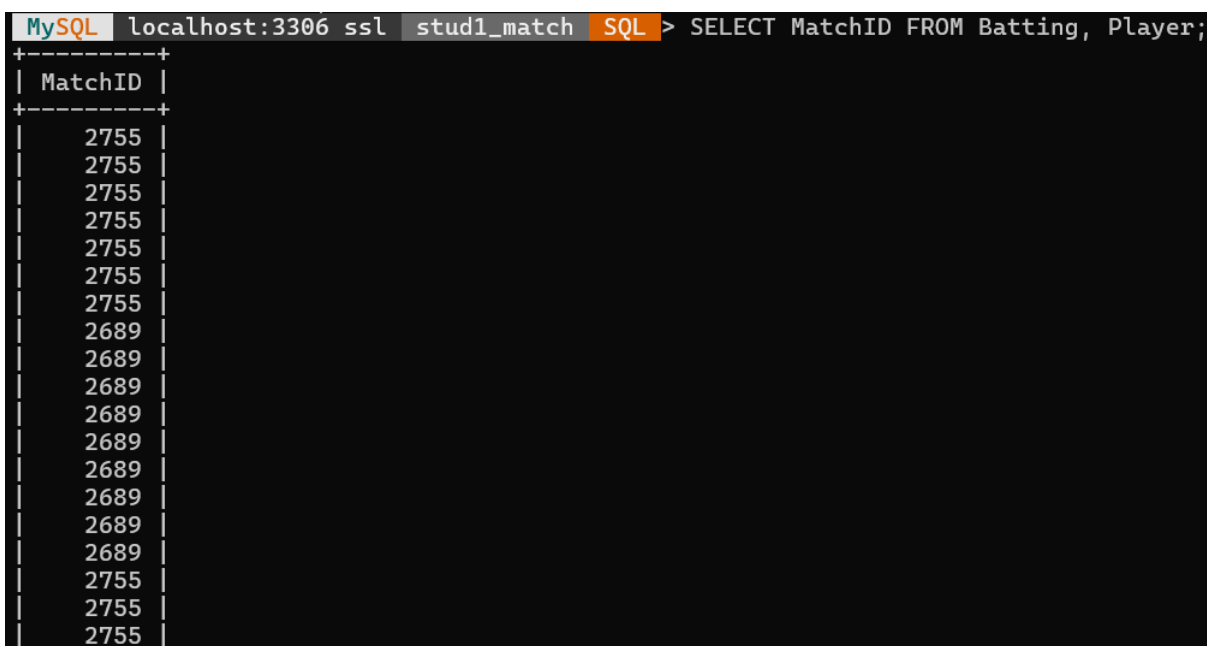
DBMS LAB 4

Parth Janardan Naukudkar_ECS_II_66_D6

Aim: To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

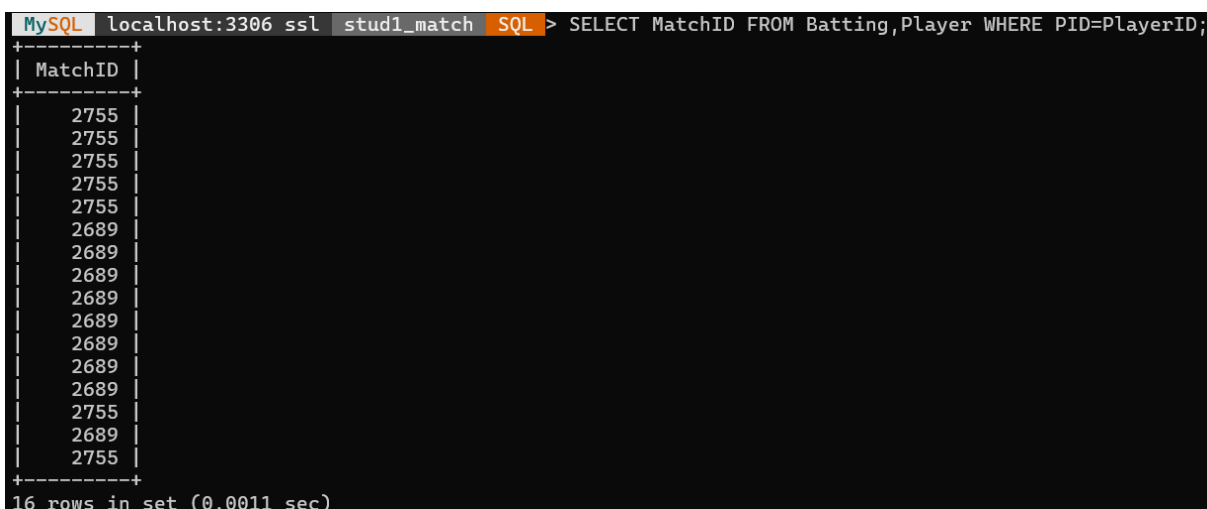
Q1. Find match Ids of all the matches in the database in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting, Player;
```



MatchID
2755
2755
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2689
2689
2755
2755
2755

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID;
```



MatchID
2755
2755
2755
2755
2755
2689
2689
2689
2689
2689
2689
2689
2689
2689
2689
2689
2755
2689
2755

16 rows in set (0.0011 sec)

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```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND  
Lname='Tendulkar';
```

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM  
Player WHERE LName='Tendulkar'); +-----+
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';  
+-----+  
| MatchID |  
+-----+  
| 2689 |  
+-----+  
1 row in set (0.0036 sec)
```

Q2. Find match information of all matches in which Dohni has batted.

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
+-----+-----+-----+-----+  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
| Pakistan | India | Peshwar | 2006-02-06 |  
+-----+-----+-----+-----+  
9280 rows in set (0.0008 sec)
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN  
(SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player  
WHERE LName ='Dhoni'));
```

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OR

Using Join

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE  
ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN (SELE  
CT PlayerID FROM Player WHERE LName ='Dhoni'));  
+-----+  
| Team1 | Team2 | Ground | Date |  
+-----+  
| Sri Lanka | India | Colombo | 2008-08-24 |  
| Australia | India | Brisbane | 2008-03-04 |  
+-----+  
2 rows in set (0.0463 sec)
```

Q3. Find match Ids of all matches in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM  
Player WHERE Lname='Tendulkar');
```

OR Using Join

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND  
LName='Tendulkar';
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');  
+-----+  
| MatchID |  
+-----+  
| 2689 |  
+-----+  
1 row in set (0.0015 sec)
```

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT  
PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);  
+-----+  
| PID |  
+-----+  
| 99001 |  
| 24001 |  
| 23001 |  
| 98002 |  
+-----+  
4 rows in set (0.0133 sec)
```

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Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 23001 |
| 89001 |
| 23001 |
| 25001 |
| 99002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
| 98002 |
+----+
11 rows in set (0.0456 sec)
```

Using Union

mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689); +-----+

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 23001 |
| 25001 |
| 99002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
+----+
9 rows in set (0.0007 sec)
```

Q6. Find Ids

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689); +-----+

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 25001 |
| 99002 |
| 95001 |
| 27001 |
+----+
5 rows in set (0.0008 sec)
```

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Q7. Find match Ids where Tendulkar has played.

```
mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID; +-----+
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0013 sec)
```

Q8. Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0007 sec)
```

Q9. Find match Ids of all those matches in which player 27001 has scored minimum runs than he made at every match he played at Brisbane

```
mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
+-----+
| MatchID |
+-----+
| 2755 |
+-----+
1 row in set (0.0007 sec)
```

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Q10(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74  
AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE  
Ground='Colombo');  
+-----+-----+  
| PID | NRuns |  
+-----+-----+  
| 25001 | 71 |  
| 91001 | 60 |  
+-----+-----+  
2 rows in set (0.0007 sec)
```

Q10(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

```
mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN  
(SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'); +-----+-----+
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Col  
ombo');  
+-----+-----+  
| PID | NRuns |  
+-----+-----+  
| 25001 | 71 |  
| 91001 | 60 |  
+-----+-----+  
2 rows in set (0.0008 sec)
```

Q11. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

```
mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PlayerID FROM Player WHERE FTest IS NULL;  
+-----+  
| PlayerID |  
+-----+  
| 27001 |  
+-----+  
1 row in set (0.0011 sec)
```

Q12. Find names of the player who batted in match 2689

```
mysql> SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM  
Batting WHERE PlayerID=PID AND MatchID=2689); OR
```

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mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND MatchID=2689;

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
```

FName	LName
Sachin	Tendulkar
Yuvraj	Singh
MS	Dhoni
Adam	Gilchrist
Ricky	Ponting
Andrew	Symonds
Breet	Lee
Praveen	Kumar
Harbhajan	Singh

9 rows in set (0.0010 sec)

Q13. Find player Ids of players who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31); +-----+

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);
```

PlayerID
24001
25001
25001
89001
91001

5 rows in set (0.0006 sec)

Q14. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground = 'Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE Batting.PID = Player.PlayerID AND Batting.MatchID = ODIMatch.MatchID));

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground = 'Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE Batting.PID = Player.PlayerID AND Batting.MatchID = ODIMatch.MatchID));
```

Empty set (0.0009 sec)

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Q15. Find the number of players that bowled in ODI match 2689

mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
+-----+
| NBowlers |
+-----+
|         4 |
+-----+
1 row in set (0.0022 sec)
```

Q16. Find the average batting score of all the players that batted in the ODI Match 2689

mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
+-----+
| AvgRuns_2689 |
+-----+
|      25.2222 |
+-----+
1 row in set (0.0012 sec)
```

Q17. Find the youngest player in the database

mysql> SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
+-----+
| Youngest_Player |
+-----+
| Ishant          |
+-----+
1 row in set (0.0009 sec)
```

Q18. Find the number of players in the database from each country

mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;

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```
MySQL localhost:3306 ssl stud1_match SQL > SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

Country	NPlayers
India	10
Australia	5
West Indies	1
Sri Lanka	3
Pakistan	2
Zimbabwe	1
South Africa	4
New Zealand	2
England	1

```
9 rows in set (0.0009 sec)
```

Q19. Find batting average of each player in the database. Present the PID of each Player.

```
mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

```
12 rows in set (0.0011 sec)
```

Solve the following

Q20. Find the batting average of each player from India in the database. Present first name of each player.

```
mysql> SELECT P.FName, SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average  
FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India'  
GROUP BY P.PlayerID, P.FName;
```

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```
MySQL localhost:3306 ssl stud1_match SQL > SELECT P.FName,
-> SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average
-> FROM Player P
-> JOIN Batting B ON P.PlayerID = B.PID
-> WHERE P.Country = 'India'
-> GROUP BY P.PlayerID, P.FName;
```

FName	Batting_Average
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

5 rows in set (0.0016 sec)

Q21. Find the average score for each player when playing against Australia

SELECT P.FName, AVG(B.NRuns) AS Average_Score FROM Player P JOIN Batting B ON P.PlayerID = B.PID JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Australia' OR M.Team2 = 'Australia' GROUP BY P.PlayerID, P.FName;

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT P.FName, AVG(B.NRuns) AS Average_Score
-> FROM Player P
-> JOIN Batting B ON P.PlayerID = B.PID
-> JOIN ODIMatch M ON B.MatchID = M.MatchID
-> WHERE M.Team1 = 'Australia'
-> OR M.Team2 = 'Australia'
-> GROUP BY P.PlayerID, P.FName;
```

FName	Average_Score
Sachin	91.0000
Yuvraj	38.0000
MS	36.0000
Adam	2.0000
Ricky	1.0000
Andrew	42.0000
Breet	7.0000
Praveen	7.0000
Harbhajan	3.0000

9 rows in set (0.0480 sec)

Q22. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

SELECT B.PID FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka' GROUP BY B.PID HAVING AVG(B.NRuns) = (SELECT MAX(AvgScore) FROM (SELECT AVG(B2.NRuns) AS

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AvgScore FROM Batting B2 JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID
WHERE M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka' GROUP BY B2.PID) AS
Averages);

```
MySQL localhost:3306 ssl stud1_match SQL > SELECT B.PID
-> FROM Batting B
-> JOIN ODIMatch M ON B.MatchID = M.MatchID
-> WHERE M.Team1 = 'Sri Lanka'
-> OR M.Team2 = 'Sri Lanka'
-> GROUP BY B.PID
-> HAVING AVG(B.NRuns) = (
->     SELECT MAX(AvgScore)
->     FROM (
->         SELECT AVG(B2.NRuns) AS AvgScore
->         FROM Batting B2
->         JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID
->         WHERE M2.Team1 = 'Sri Lanka'
->         OR M2.Team2 = 'Sri Lanka'
->         GROUP BY B2.PID
->     ) AS Averages
-> );

+-----+
| PID   |
+-----+
| 25001 |
+-----+
1 row in set (0.0020 sec)
```